

carbo::lyser™ II / III - V3

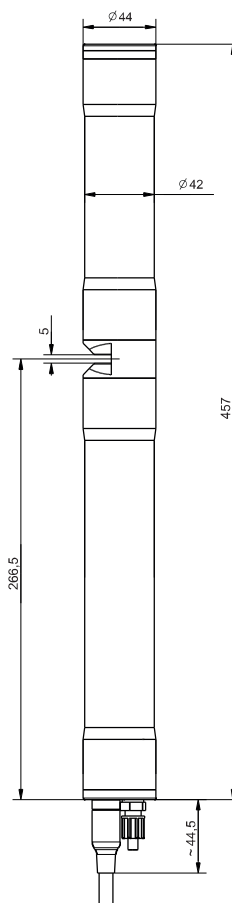
carbo::lyser™ II monitors 2 of the following parameters: TSS, TS, TOC, DOC, BOD, COD, COD f, UV254 and UV254 f

carbo::lyser™ III monitors 3 of the following parameters: TSS, TOC, DOC, BOD, COD, COD f, UV254 and UV254 f

- measuring principle: UV-Vis spectrometry over the total range (190-750 nm)
- web server on board - IoT enabled, no user software is needed to configure the probe
- communicates directly with your mobile device via WLAN
- 8 GB onboard memory - capacity for logging data for many years
- improved optical performance - revolutionary precision
- fast measurement interval - every 10 seconds possible
- extremely power efficient - sleep mode for low energy consumption
- multiparameter probe with 1 mm, 5 mm or 35 mm optical path length, ideal for waste water, surface water and drinking water
- long term stable and maintenance free in operation
- factory precalibrated, local multi-point calibration possible
- automatic cleaning with compressed air or brush

recommended accessories

part number	article name
B-32-xxx	s::can compressor
B-33-012	con::nect V3
B-44	cleaning valve
B-44-2	
C-32-V3	Adapter cable to connect a V3 spectrometer (M12) to V2 Terminal (MIL Plug)
D-330-xxx	con::cube V3
F-110-V3	carrier s::can spectrometer V3 & V2 probe, 45°
F-120-V3	carrier s::can spectrometer V3 & V2 probe, vertical attachment
F-48-V3	spectrometer V3 & V2 flow-cell (bypass setup), PVC
S-11-xx-moni	moni::tool Software



technical specification

measuring principle	UV-Vis spectrometry 190 - 750 nm
measurement interval	10 sec (configurable, depending on application)
automatic compensation cross sensitivities	turbidity / solids / organic substances
precalibrated ex-works	all parameters
accuracy standard solution (>1 mg/l)	NO ₃ -N: +/- 3% +1/OPL[mg/l]* COD-KHP: +/-3% +10/OPL[mg/l]* (* OPL ... optical pathlength in mm)
access to raw signals	no
reference standard	distilled water
onboard memory	8 GB
integrated temperature sensor	0 ... 45 °C
resolution temperature sensor	0.1 °C
integration via	con::cube V3 con::nect V3 con::lyte V5 (D-320-pro2) and adapter cable (C-32-V3)
power supply	10 ... 18 VDC
power consumption (typical)	3 W
power consumption (sleep model)	60 mW
power consumption (max.)	20 W
interface to s::can terminals	M12 RSTS 8Y (IP67), RS485, Ethernet
interface to third party terminals	con::nect V3 incl. Modbus RTU, REST API, Modbus TCP/IP
digital interface (for cleaning devices)	1 digital in/out 1 digital out
network connection	100Base-T Ethernet, WLAN
status information	RGB LED ring
internal sensors	supply voltage sensor, tilt sensor, rotation sensor

cable length	1 m fixed cable (-010) or 7.5 m fixed cable (-075) or 15 m fixed cable (-150)
cable type	PU jacket
housing material	stainless steel 1.4404
window material	optical path length 5 and 1 mm: sapphire optical path length 35 mm: fused silica (UV-grade)
weight (min.)	3.4 kg (incl. cable)
dimensions (Ø x l)	optical path length 35 mm: 44 x 473 mm / 517.5 mm optical path length 5 mm: 44 x 457 mm / 501.5 mm optical path length 1 mm: 44 x 453 mm / 497.5 mm
operating temperature	0 ... 45 °C
operating pressure	0 ... 3 bar
high pressure specification (optional)	10 bar
installation / mounting	submersed or in a flow cell
flow velocity	3 m/s (max.)
mechanical stability	30 Nm
ingress protection class	IP68
automatic cleaning	media: compressed air or autobrush permissible pressure: 3 ... 6 bar
storage temperature	-10 ... 65 °C
conformity - environmental testing	EN 60721-3
conformity - EMC	EN 61326-1
conformity - RoHS 2	EN 50581
standard warranty	2 years
extended warranty (optional)	3 years

municipal WWTP influent & sewer

		parameter								part number
		TSS [mg/l]	TOC [mg/l]	DOC [mg/l]	BOD [mg/l]	COD [mg/l]	COD f [mg/l]	UV254 [Abs/m]	UV254 f [Abs/m]	
carbo::lyser™ II (2 parameters, 1 mm OPL)	min.	0	0	0	0	0	0	0	0	G3-C2-I-01-NO-xxx
	max.	8000	3300	2600	5300	10000	5300	3300	2800	
carbo::lyser™ II (2 parameters, 5 mm OPL)	min.	0	0	0	0	0	0	0	0	G3-C2-I-05-NO-xxx
	max.	1200	500	400	800	1500	800	500	420	
carbo::lyser™ III (3 parameters, 1mm OPL)	min.	0	0	0	0	0	0	0	0	G3-C3-I-01-NO-xxx
	max.	8000	3300	2300	5300	10000	5300	3300	2800	
carbo::lyser™ III (3 parameters, 5 mm OPL)	min.	0	0	0	0	0	0	0	0	G3-C3-I-05-NO-xxx
	max.	1200	500	400	800	1500	800	500	420	

municipal WWTP effluent

		parameter								part number
		TSS [mg/l]	TOC [mg/l]	DOC [mg/l]	BOD [mg/l]	COD [mg/l]	COD f [mg/l]	UV254 [Abs/m]	UV254 f [Abs/m]	
carbo::lyser™ II (2 parameters, 5 mm OPL)	min.	0	0	0	0	0	0	0	0	G3-C2-E-05-NO-xxx
	max.	600	400	300	300	500	300	500	420	
carbo::lyser™ III (3 parameters, 5 mm OPL)	min.	0	0	0	0	0	0	0	0	G3-C3-E-05-NO-xxx
	max.	600	400	300	300	500	300	500	420	

ammo::lyser™ eco

ammo::lyser™ II eco: monitors $\text{NH}_4\text{-N}$ and temperature

ammo::lyser™ III eco+pH additionally monitors pH

ammo::lyser™ III eco+ $\text{NO}_3\text{-N}$ additionally monitors $\text{NO}_3\text{-N}$

ammo::lyser™ III eco+ Cl^- additionally monitors chloride

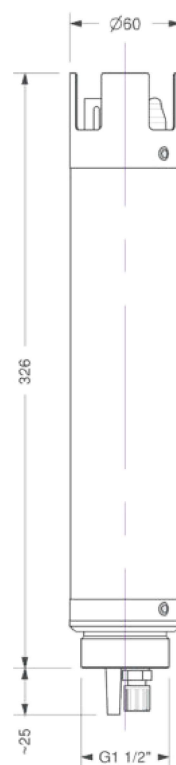
ammo::lyser™ IV eco+pH+ $\text{NO}_3\text{-N}$ additionally monitors pH and $\text{NO}_3\text{-N}$

ammo::lyser™ VI eco+pH+ Cl^- additionally monitors pH and chloride

- s::can plug & measure
- measuring principle: ISE (ionselective electrodes) - without potassium compensation
- multiparameter probe
- long term stable, factory precalibrated
- minimal maintenance, automatic cleaning with compressed air
- unique, non-porous / non-leaking reference electrode for technically unrivalled and consistent performance
- ISE refurbishment - the easy way to minimise maintenance
- easy & quick mounting and measurement directly in the media (InSitu) or in a flow cell (monitoring station)
- automatic temperature compensation and pH compensation possible
- ideal for surface water, ground water, drinking water and waste water
- life time of ISE: typically 6 month (for applications $<1\text{mg/l}$ $\text{NH}_4\text{-N}$), resp. 1 to 2 years (for applications $>1\text{mg/l}$ $\text{NH}_4\text{-N}$)
- plug connection or fixed cable

recommended accessories

part number	article name
B-44	cleaning valve
B-44-2	
C-210-sensor	10 m extension cable for s::can physical probes and s::can ISE probes
F-11-oxi-ammo	carrier oxi::lyser / soli::lyser / s::can ISE probes
F-48-ammo	ammo::lyser flow-cell (by-pass setup), PVC



technical specification

measuring principle	ISE	power consumption (typical)	0.72 W
measuring principle detail	NH ₄ -N: ionophore membrane pH: non-porous reference electrode NO ₃ -N: ionophore membrane Cl ⁻ : ionophore membrane	interface to s::can terminals	sys plug (IP67), RS485
resolution	NH ₄ -N, K, NO ₃ -N, Cl, F: 0.01 at 0.02 ... 19.99 mg/l 0.1 at 20.0 ... 99.9 mg/l 1 at 100 ... 1000 mg/l T: 0.1 °C	cable length	7.5 m fixed cable (-075) or plug connection (-000)
accuracy (standard solution)	NH ₄ -N: +/-3% or +/-0.5mg/l* (*whichever is greater)	cable type	PU jacket
automatic compensation cross sensitivities	E-532-eco-xxx: temp E-532-eco-pH-xxx: temp, pH E-532-eco-NO ₃ -N-xxx: temp E-532-eco-NO ₃ -N-pH-xxx: temp, pH E-532-eco-CL-xxx: temp E-532-eco-CL-pH-xxx: temp, pH	housing material	stainless steel 1.4571, POM-C
precalibrated ex-works	all parameters	weight (min.)	2.7 kg
response time (T90)	0 ... 60 sec.	dimensions (Ø x l)	60 x 326 mm
integration via	con::cube con::lyte con::nect	operating temperature	0 ... 60 °C
power supply	10 ... 30 VDC	operating pressure	0 ... 1 bar
		installation / mounting	submersed or in a flow cell
		process connection	bayonet
		flow velocity	0.01 m/s (min.), 3 m/s (max.)
		automatic cleaning	media: compressed air permissible pressure: 2 ... 4 bar
		storage temperature (electrode)	2 ... 40 °C
		storage temperature (sensor)	2 ... 40 °C
		conformity - EMC	EN 50081-1, EN 50082-1, EN 60555-2, EN 60555-3
		conformity - safety	EN 61010-1
		protection class (-000)	IP67
		protection class (-075)	IP68

measuring range

		parameter				part number
		NH ₄ -N [mg/l]	NO ₃ -N [mg/l]	pH [pH]	temperature [°C]	
ammo::lyser™ II eco (NH ₄ -N, temp)	min.	0.1			0	E-532-eco-000 / -075
	max.	1000			60	
ammo::lyser™ III eco+NO ₃ -N (NH ₄ -N, temp, NO ₃ -N)	min.	0.1	0.3		0	E-532-eco-NO ₃ -N-000 / -075
	max.	1000	1000		60	
ammo_lyser_III_eco_pH (NH ₄ -N, Temp, pH)	min.	0.1		2	0	E-532-eco-pH-000 / -075
	max.	1000		12	60	
ammo::lyser™ IV eco+NO ₃ -N+pH (NH ₄ -N, temp, NO ₃ -N, pH)	min.	0.1	0.3	2	0	E-532-eco-NO ₃ -N-pH-000 / -075
	max.	1000	1000	12	60	

Spectrometer
Probes

i::scan

Intelective
Probes

Physical Probes

Terminals

Software

System
Configuration

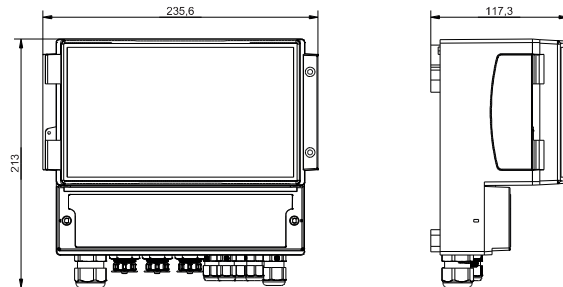
Monitoring
Stations

Spare Parts &
Accessories

Services &
Solutions

con::lyte

- low-cost terminal for control applications
- power efficient LCD display and ergonomic UI
- sensor and station management of up to 2 (eco) or 6 (pro) parameters
- control of automatic cleaning, data logging, sample & calibration, sensor function check and easy data transfer via USB-stick
- process interface to SCADA or con::cube via Modbus RTU, Profibus DP, analog 4-20 mA and relay outputs (state/PWM/Pulse)
- integration of third-party sensors via analog and digital I/Os
- outstanding control features: easy threshold and alarm limits with hysteresis, 3 opt. PID or 2-point controllers
- certifications: CE, UL, CSA and RCM



technical specification

display	LCD
function indicator	2 x LED
operation via	keypad
onboard memory	512 MB
interface to SCADA	Modbus RTU (optional), Profibus DP (optional), analog outputs
data transfer	USB stick
power supply	100-240 VAC (50-60 Hz)
power consumption (max.)	25 W
analog inputs	1 x 0/4-20 mA
outputs for automatic cleaning	1 (2nd cleaning device via relay output)
digital inputs	2
digital input flow detector	1
relay outputs	2 x 6A (600 VAC)
system error relay	1 x 6A (600 VAC)
dimensions (width x height x depth)	235.6 x 213 x 117.3 mm
housing material	PC
weight (min.)	1300 g
operating humidity	5 ... 90 %
storage temperature	-20 ... 50 °C
storage humidity	5 ... 90 %
ingress protection class	IP65
conformity - EMC	EN 61326-1
conformity - safety	EN 61010-1
conformity - RoHS 2	EN 50581

con::lyte eco (2 parameters)

integration of	1 x i::scan, s::can sensor or s::can ISE probe
interface to s::can sensors	1 x sys plug, RS485
analog outputs	2 x 4-20 mA
operating temperature (eco)	-20 ... 45 °C
part number 230V	D-320-eco-230

con::lyte pro (6 parameters)

integration of	pro1: i::scan, s::can sensors/ISE probes; pro2: s::can G::series, i::scan, s::can sensors/ISE probes
interface to s::can spectrometric probes	D320-pro2: 1 x MIL, RS485
interface to s::can sensors	D-320-pro1: 3 x sys plug, RS485 D-320-pro2: 2 x sys plug, RS485
analog outputs (optional license)	3 x 4-20 mA
analog outputs (optional module)	2 x 4-20 mA / 4 x 4-20 mA
operating temperature (pro1)	-20 ... 45 °C
operating temperature (pro2)	-20 ... 50 °C
part number 230V	D-320-pro1-230, D-320-pro2-230

recommended accessories

part number	article name
C-31-eu	Optional 2 m power cable
C-32-V3	Adapter cable to connect a V3 spectrometer (M12) to V2 Terminal (MIL Plug)
D-319-logger	Datalogger option for con::lyte
D-319-out-profibus	Profibus (output module for con::lyte)
D-320-PID	3 x PID control output for con::lyte D-320
D-320-out-mA	License for 3 analog outputs (4-20 mA) for con::lyte pro